# Machine Learning for Economists and Business Analysts

Anthony Strittmatter

# Lecturer

**Anthony Strittmatter**

**Research Interests:** Business, Labour, and Health Economics, Program Evaluation, Computational Data Analytics

**Positions:**

| | |
|---|---|
| Since 2023 | Full Professor for Applied Econometrics at UniDistance Suisse in Brig/Valais |
| 2022-2024 | Senior Economist at Amazon in London |
| 2020-2023 | Institut Polytechnique and CREST in Paris |
| 2014-2020 | University of St.Gallen, with research visits at UC Berkeley, Stanford University, and LMU Munich |
| 2009-2016 | Albert-Ludwig University of Freiburg |

- Email: anthony.strittmatter@unidistance.ch
- Webpage: www.anthonystrittmatter.com

# Course Outline

- **Lecture 1:** Introduction to Statistical Learning
  - Out-of-Sample Prediction
  - Prediction vs. Causal Analysis
- **Lecture 2:** Regularized Regression
  - Lasso, Ridge, Elastic Net
- **Lecture 3:** Non-parametric Supervised Machine Learning
  - Classification and Regression Trees
  - Random Forests
  - Deep Learning

# Course Outline (cont.)

- **Lecture 4:** Unsupervised Machine Learning
  - Clustering
  - Principal Component Analysis
- **Lecture 5:** High-Dimensional Confounding
  - Double Selection Procedure
  - Double Machine Learning
- **Lecture 6:** Effect Heterogeneity and Policy Learning
  - Causal Forest
  - Optimal Policy Learning
  - Bandit Algorithms

# PC Labs

- PC labs are integral part of the course.
- I will provide the entire course material on my GitHub repository: https://github.com/AStrittmatter/Machine-Learning-Course
- I will use interactive Jupyter notebooks during the PC labs: https://mybinder.org/v2/gh/AStrittmatter/Machine-Learning-Course/HEAD
- The advantage of the notebooks is that you do not need to install anything and the data is in the correct folder.
- However, in case the connection to the server is weak I also provide an R-file for download on my GitHub repository. To use it, you need to install R and RStudio on your PC. This are both open source softwares.

# Grading

- Group Data Challenge (Deadline for submission: March 24, 2026)
- Individual Home Assignment (Deadline for submission: April 26, 2026)
- Grades: Pass/Fail

# General

- Feel free to interupt me at any time when you have questions.
- Tell me when I'm too slow or too fast. Ask me to repeat material in case something was not clear.
- You can also send me an email with questions: anthony.strittmatter@unidistance.ch
- Proposals to improve the course are also welcome.
- Please try to interact as much as possible with your fellow students. Build learning groups.

# References


- Mullainathan and Spiess (2017): "Machine Learning: An Applied Econometric Approach", Journal of Economic Perspectives, 31 (2), pp. 87-106, download.
- Athey (2017): "Beyond Prediction: Using Big Data for Policy Problems", Science, 355 (6324), pp. 483-485, download.

# What is Machine Learning (ML)?

- ML (or statistical learning) methods exist already since decades.
- Currently "Machine Learning" is a buzz word
- Probably most people think of ML as some computational intensive methods that make data-driven modelling decisions and/or can deal with large data amounts.
- However, relevant textbooks consider even OLS/Logit as a statistical learning tool.

# Purpose of Machine Learning

- Consider the structural model

$$Y = f(X) + \epsilon = X\beta + \epsilon,$$

  with $E[\epsilon] = 0$.
- Causal analysis has the purpose to estimate $\hat{\beta}$, with $plim(\hat{\beta}) = \beta$.
- **Machine learning** has the purpose to predict $Y$.
- There is a clear link between causal analysis and machine learning, because

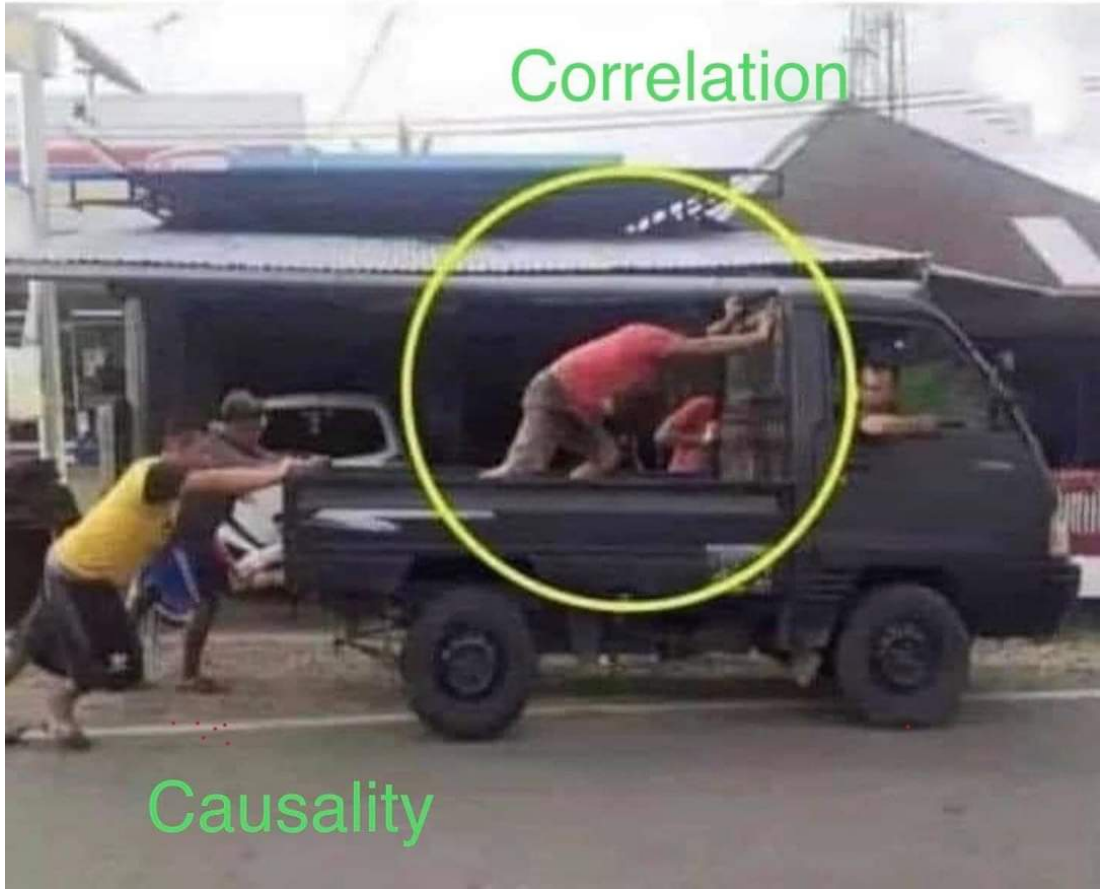
$$\widehat{Y} = \hat{f}(X) = X\hat{\beta}$$

  is a potential predictor for $Y$.
- Parameter consistency has not the highest priority when it comes to predictions.

# Potential Advantages and Disadvantages of ML

- ML methods can be very powerful to predict $Y$, even when $\hat{\beta}$ is biased.
- ML methods can incorporate many (or even high-dimensional) covariates $X$ in a convenient way.
- ML methods can model $\hat{f}(\cdot)$ in a very flexible and data-driven way.
- **Main disadvantage:** ML is a black-box approach and we loose the interpretability of $\hat{f}(\cdot)$ or $\hat{\beta}$.

# Prediction vs. Causality



# Causal vs. Predictive Questions

**Predictive Questions:**

- How will the oil price change tomorrow (forecasting)?
- How high is the current unemployment rate (nowcasting)?
- Which adolescents have a high probability of becoming addicted to drugs (policy prediction)?

**Causal Questions:**

- What is the effect of a tweet by president Donald Trump on oil prices?
- How does inflation affect the unemployment rate?
- Can prevention programs reduce the probability of drug addiction among high risk youths?

# Assessing the Model Accuracy

**Causal Analysis:**

- True $\beta$ is unobservable.
- Assess the model with asymptotic properties

$$\sqrt{N}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \sigma^2).$$

- Finite sample biases are mostly neglected.

# Assessing the Model Accuracy

**Prediction:**

- We observe $Y$ for each unit (e.g. individual).
- We can assess the model accuracy directly in the sample of our analysis, for example, using the mean-squared-error (MSE)

$$\frac{1}{N}\sum_{i=1}^{N}(Y_i - \hat{Y}_i)^2.$$

- MSE accounts for finite sample biases.

# Example: Prediction of Used Car Prices

- We have access to web-scraped data from the online advertisement platform *myLemons*.
- We want to predict asking prices of used cars based on observable characteristics.
- We observe around 40 covariates about car brand, mileage, age, emissions, maintenance certificate, seller type, guarantee, etc. (including several non-linear and interaction terms)

# In-Sample MSE

- Partition data into training and test sample
- In the training sample, we estimate the empirical model

$$Y_{tr} = \hat{f}_{tr}(X_{tr}) + \hat{\epsilon}_{tr} = X_{tr}\hat{\beta}_{tr} + \hat{\epsilon}_{tr}$$

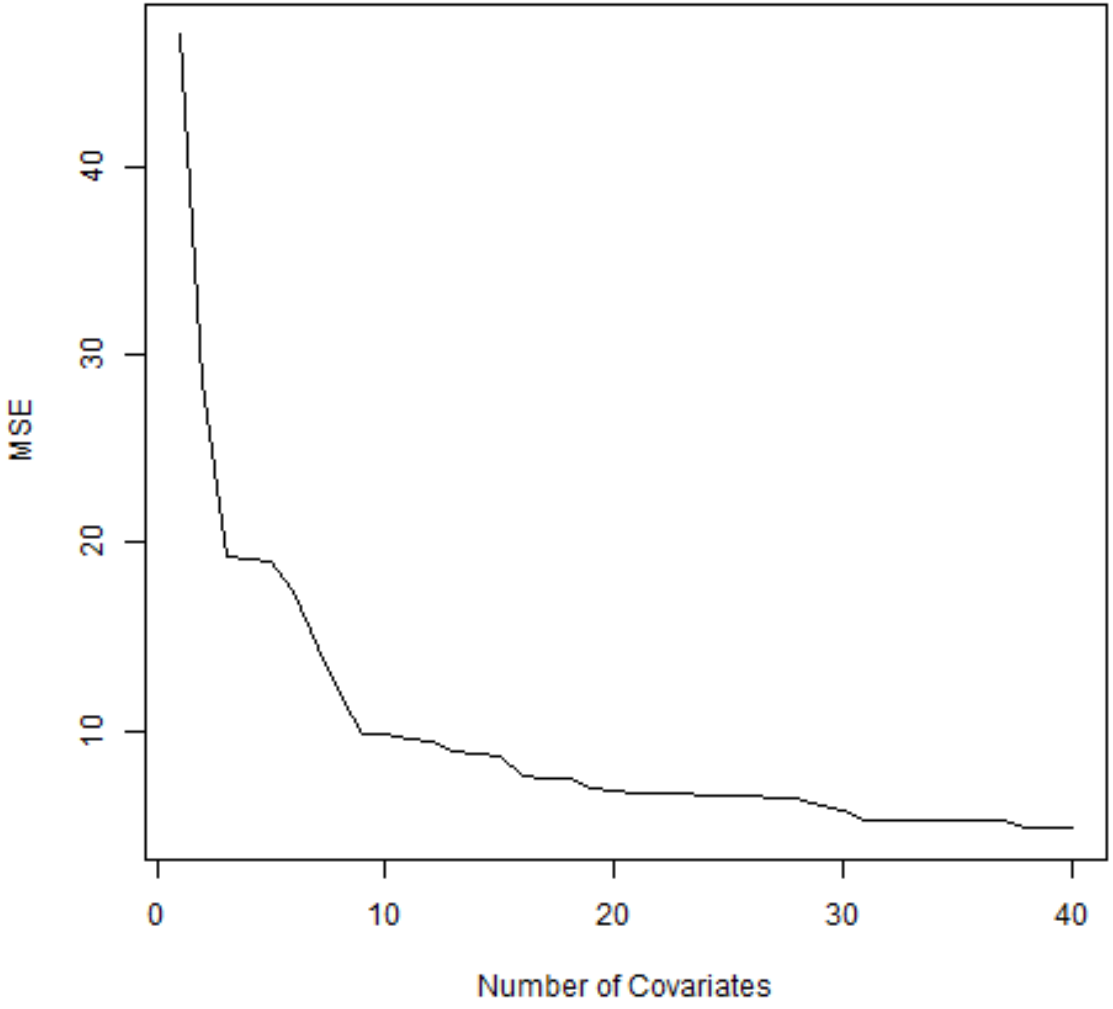
- In the training sample, we predict the fitted values

$$\widehat{Y}_{tr} = \hat{f}_{tr}(X_{tr}) = X_{tr}\hat{\beta}_{tr}$$

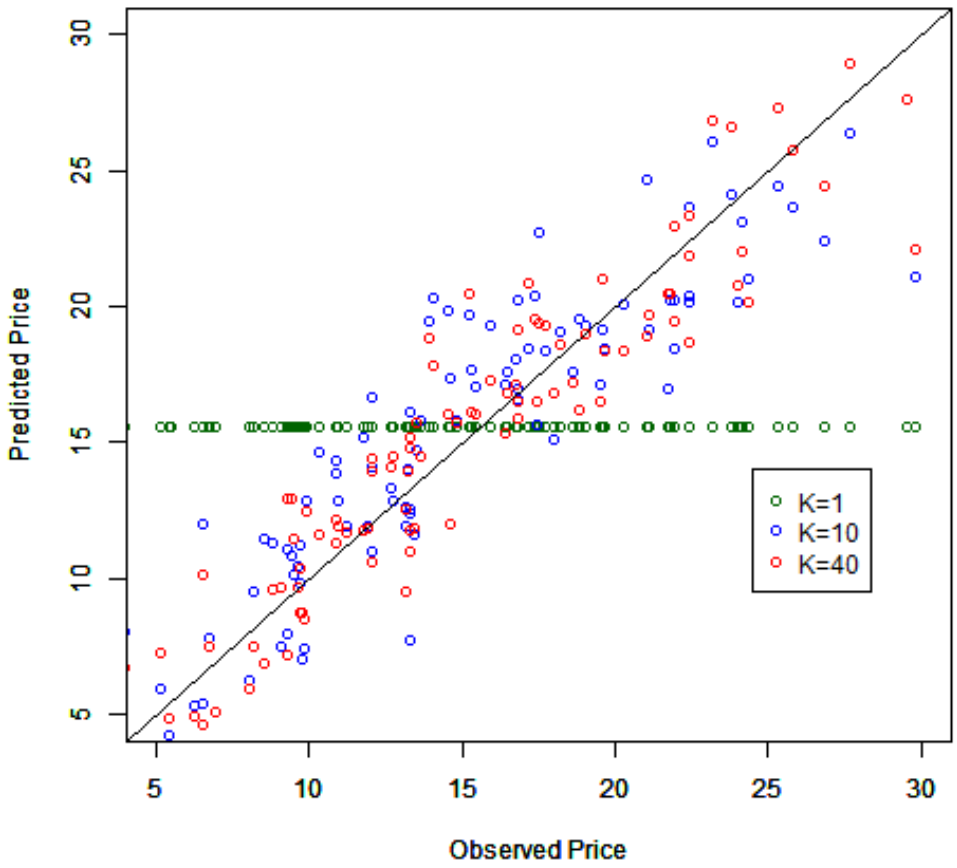
  and calculate the MSE

$$\widehat{MSE}_{tr} = \frac{1}{N_{tr}} \sum_{i=1}^{N_{tr}} (Y_{i,tr} - \widehat{Y}_{i,tr})^2.$$

# MSE in Training Sample



# Predicted Car Prices in Training Sample




| Number of Covariates | 1 | 10 | 40 |
|---|---|---|---|
| MSE | 46.948 | 9.819 | 4.866 |

# Out-of-Sample MSE

- In the training sample, we estimate the empirical model

$$Y_{tr} = \hat{f}_{tr}(X_{tr}) + \hat{\epsilon}_{tr} = X_{tr}\hat{\beta}_{tr} + \hat{\epsilon}_{tr}$$

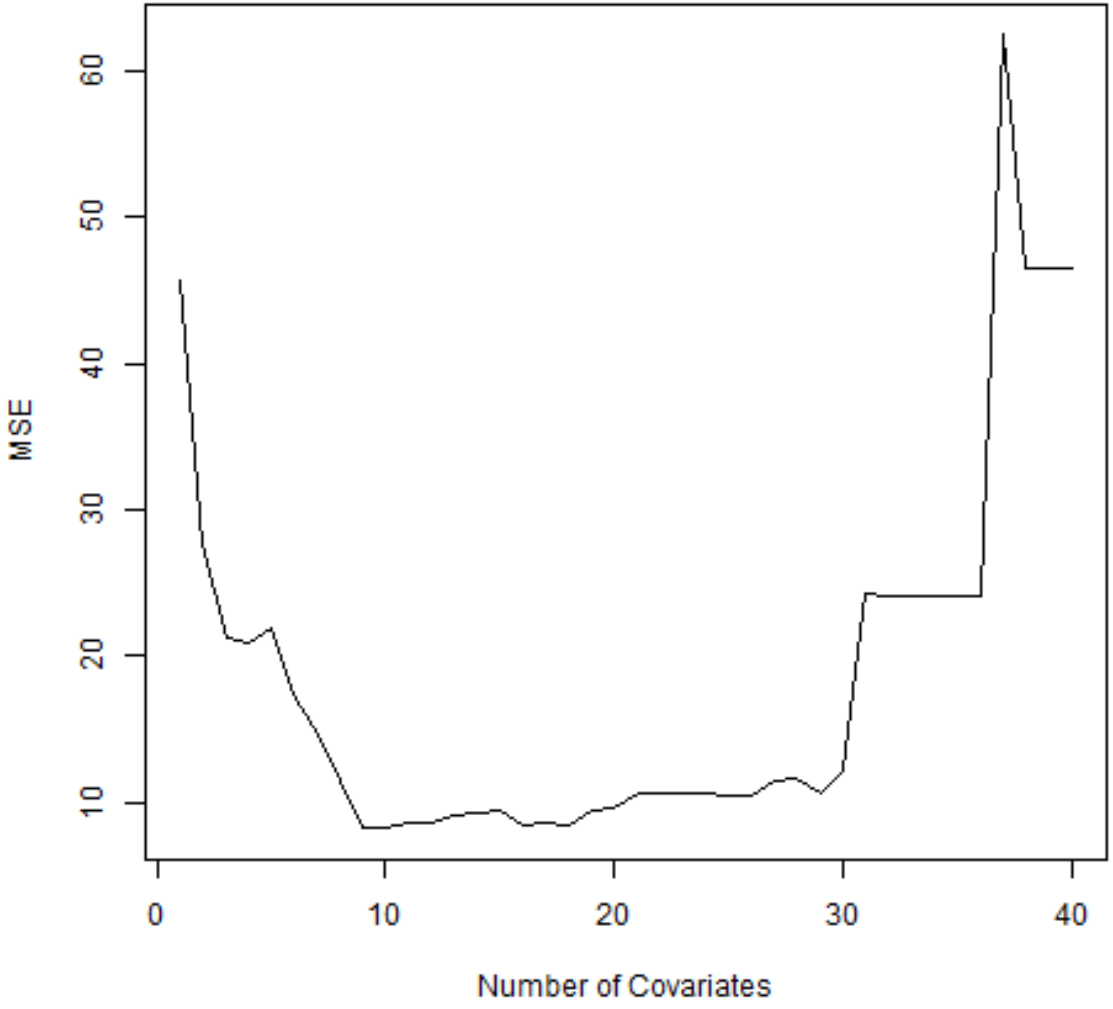
- In the test sample, we predict the fitted values

$$\widehat{Y}_{te} = \hat{f}_{tr}(X_{te}) = X_{te}\hat{\beta}_{tr}$$

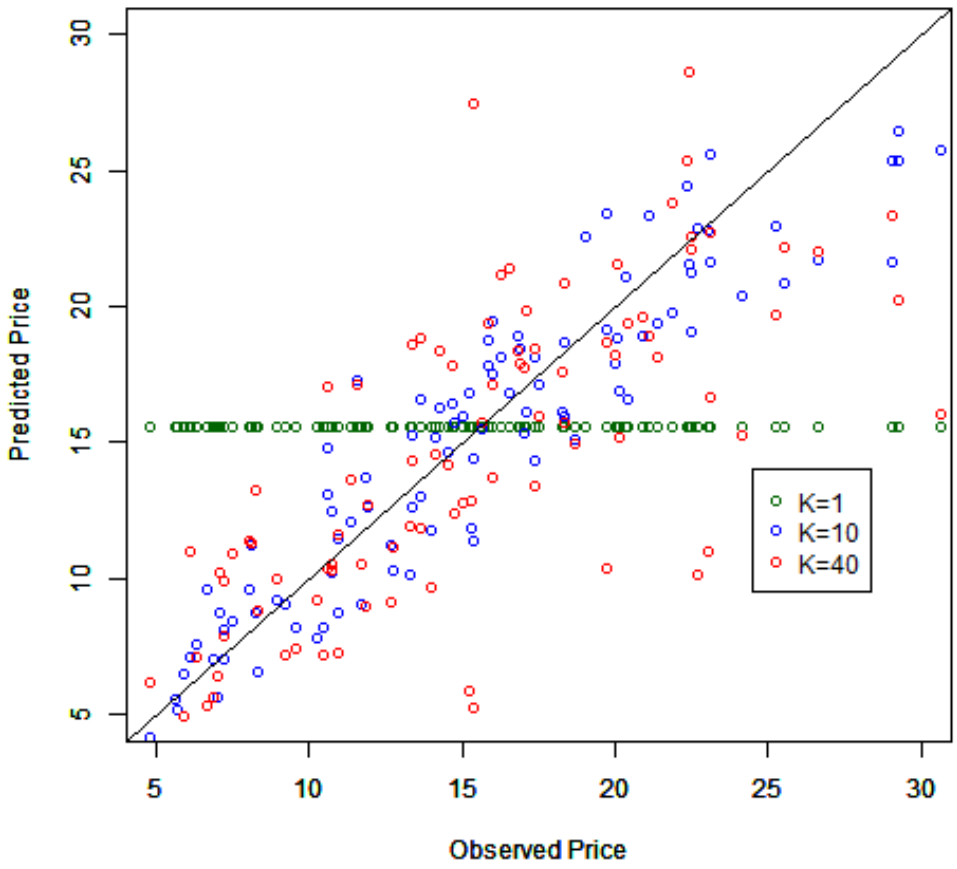
  and calculate the MSE

$$\widehat{MSE}_{te} = \frac{1}{N_{te}} \sum_{i=1}^{N_{te}} (Y_{i,te} - \widehat{Y}_{i,te})^2.$$

# MSE in Test Sample



# Predicted Car Prices in Test Sample




| Number of Covariates | 1 | 10 | 40 |
|---|---|---|---|
| MSE | 45.742 | 8.222 | 46.499 |

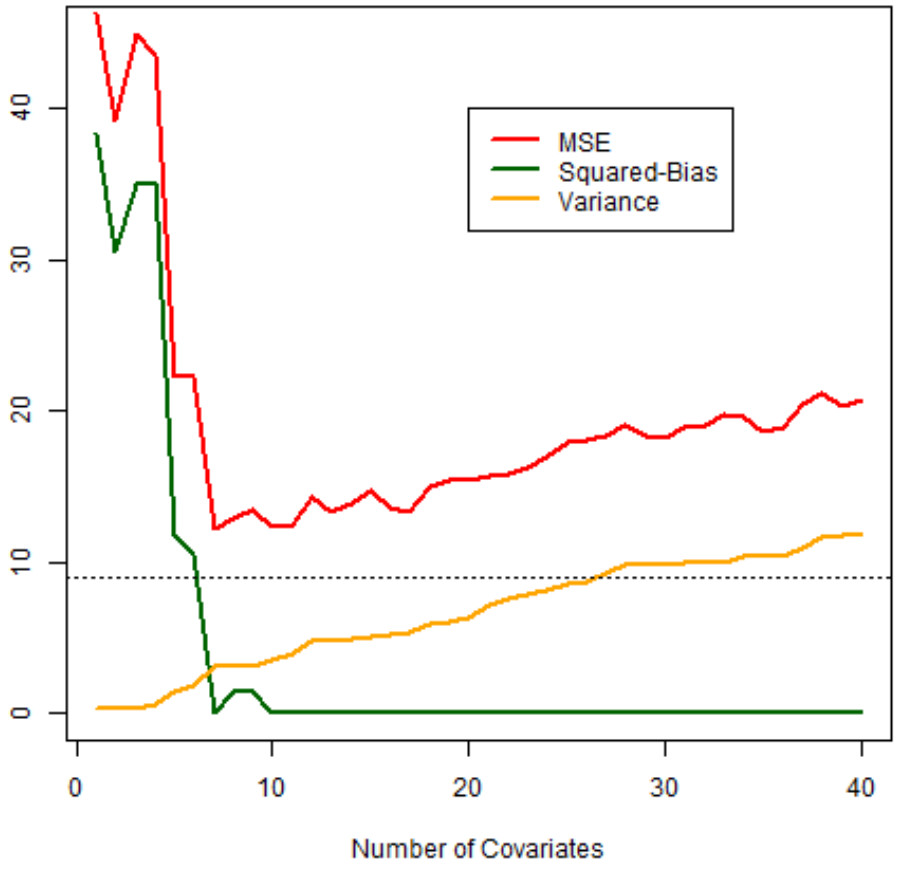
# Bias-Variance Trade-Off

- When we assess the model for one randomly drawn individual from the test sample with fixed characteristics $x_{te}$, then we can decompose the MSE to

$$
\begin{aligned}
MSE_{te} =& E[(Y_{te} - \widehat{Y}_{te})^2] \\
=& E[(f(x_{te}) + \epsilon_{te} - \hat{f}_{tr}(x_{te}))^2] \\
=& \underbrace{E[(f(x_{te}) - \hat{f}_{tr}(x_{te}))^2]}_{\text{Reducible}} + \underbrace{Var(\epsilon_{te})}_{\text{Irreducible}} \\
=& \underbrace{E[f(x_{te}) - \hat{f}_{tr}(x_{te})]^2}_{\text{Squared-Bias}} + \underbrace{Var(\hat{f}_{tr}(x_{te}))}_{\text{Variance}} + Var(\epsilon_{te})
\end{aligned}
$$

- For i.i.d. data, $\hat{f}_{tr}(\cdot)$ and $\epsilon_{te}$ are independent of each other.

# Simulation of Bias-Variance Trade-Off




- Only the first ten covariate have an impact on car prices in the simulation.
- Horizontal dashed line is the simulated noise $Var(\epsilon_{te})$.

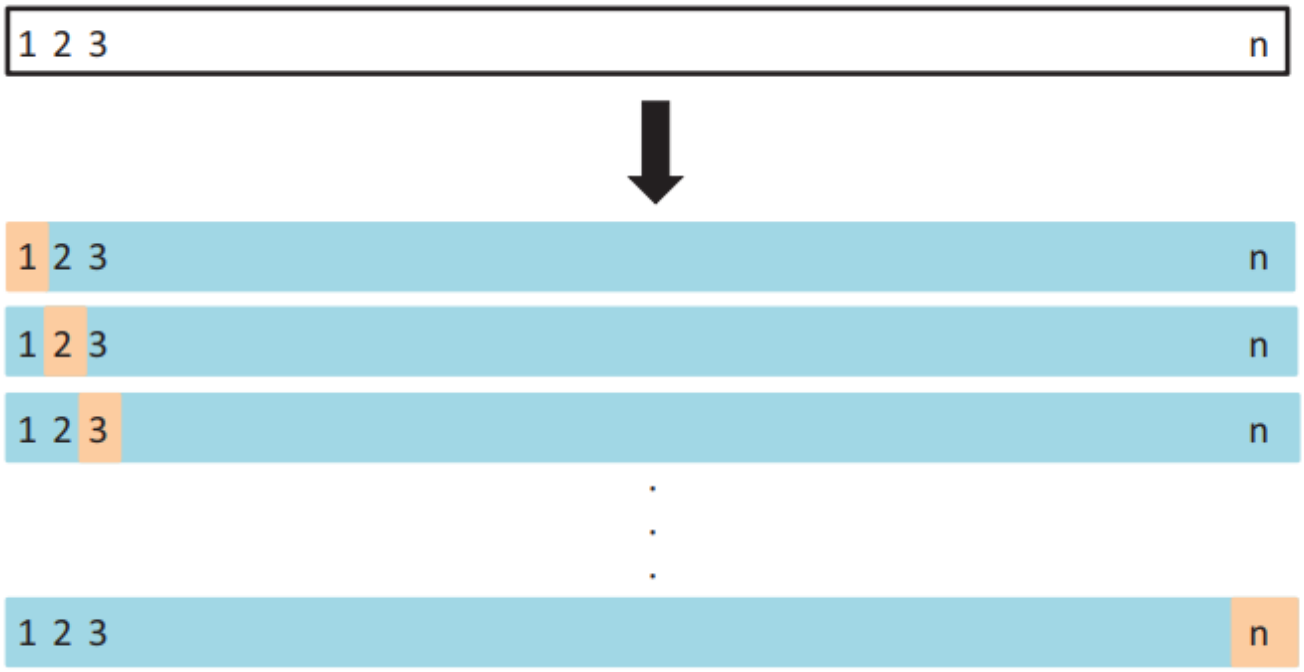
# Lasso Example

$$\arg\min_{\beta} \left\{ \sum_{i=1}^{N} \left( Y_i - \beta_0 - \sum_{j=1}^{p} X_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}$$

| | OLS | Lasso |
|---|---|---|
| Intercept | 21.246 | 22.776 |
| diesel | 2.075 | . |
| other_car_owner | 0.730 | . |
| pm_green | 1.635 | . |
| private_seller | 6.100 | 0.076 |
| guarantee | -2.440 | -0.437 |
| inspection | -0.813 | . |
| maintenance_cert | 1.481 | . |
| mileage | -0.049 | -0.031 |
| age_car_years | -1.291 | -1.012 |
| $R^2$ training | 0.655 | 0.543 |
| $R^2$ test | 0.606 | 0.611 |

# Selection of Optimal Penalty Parameter

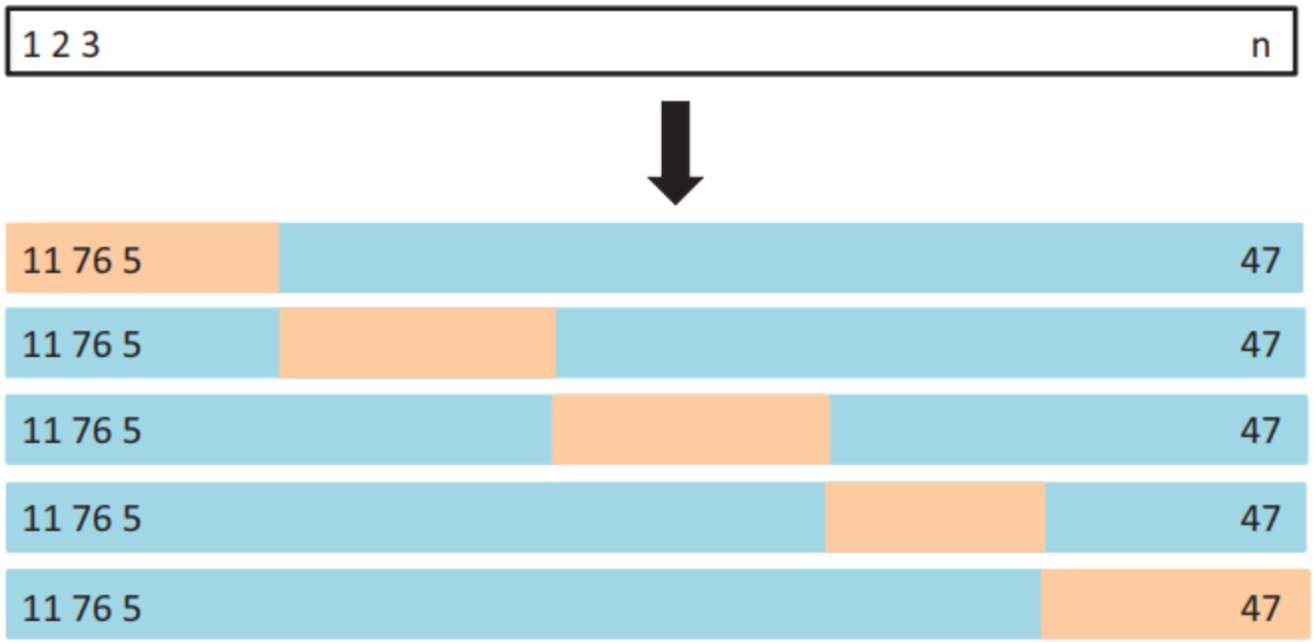
**Leave-One-Out Cross-Validation**




Source: James et al. (2013), p. 179

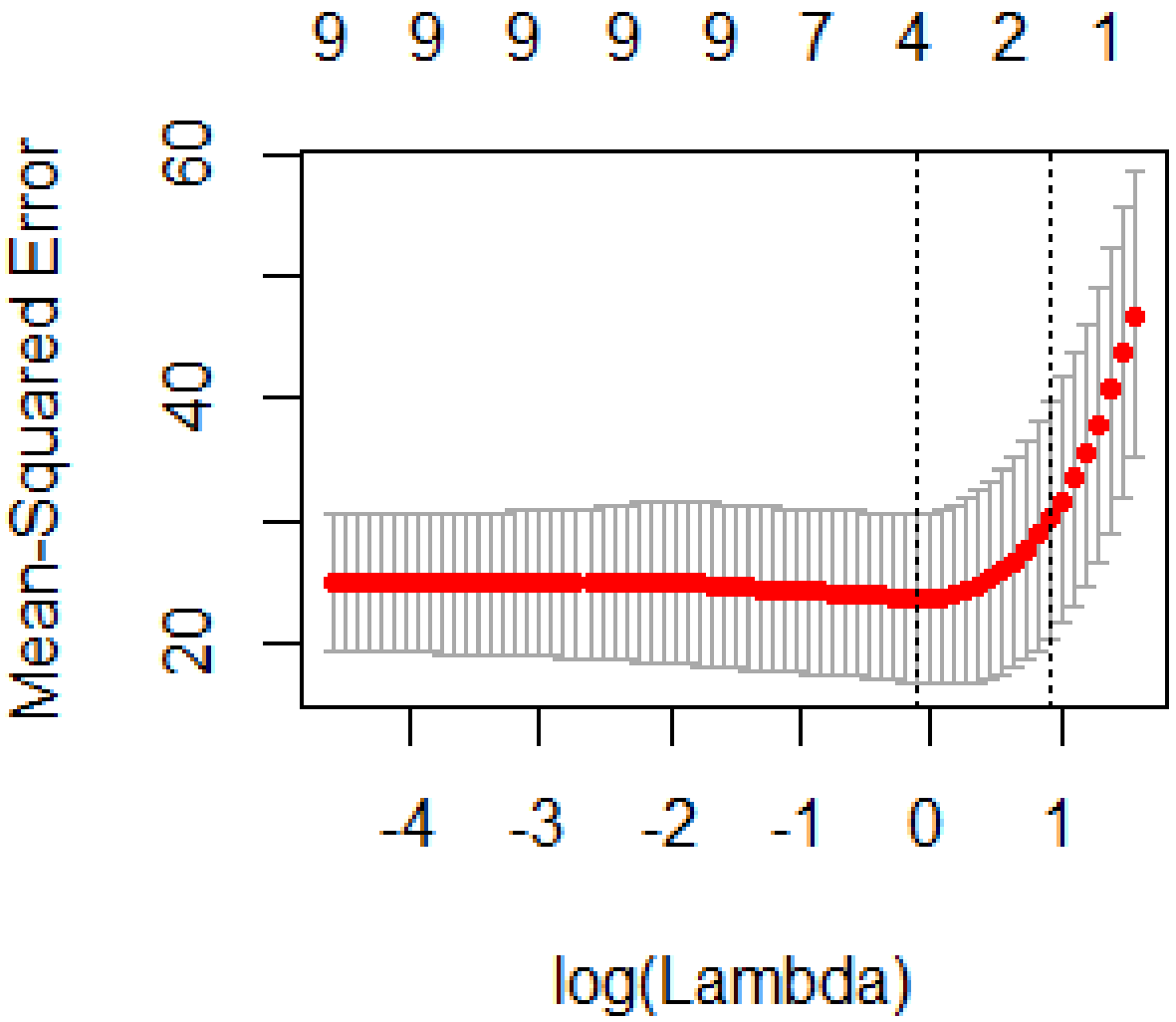
# Selection of Optimal Penalty Parameter

**k-fold Cross-Validation (CV) Algorithm**




Source: James et al. (2013), p. 181

# Cross-Validated MSE



# Stability of the Lasso Model

| | Lasso 1 | Lasso 2 | Lasso 3 | Lasso 4 | Lasso 5 |
|---|---|---|---|---|---|
| Intercept | 22.776 | 25.947 | 24.937 | 27.309 | 25.116 |
| diesel | . | . | 2.387 | . | 0.886 |
| other_car_owner | . | -1.257 | 0.393 | . | . |
| pm_green | . | 2.871 | . | . | . |
| private_seller | 0.076 | 5.094 | . | -1.037 | . |
| guarantee | -0.437 | 1.677 | 15.939 | . | . |
| inspection | . | -0.666 | -0.374 | . | . |
| maintenance_cert | . | -2.579 | -0.868 | . | . |
| mileage | -0.031 | -0.037 | -0.041 | -0.069 | -0.062 |
| age_car_years | -1.012 | -1.347 | -1.416 | -0.874 | -1.115 |

→ We do not learn the "true" structural model from ML

→ ML is a black-box approach

# Stability of the Lasso Predictions

**Correlation of Predicted Car Prices in Test Sample:**

| | Lasso 1 | Lasso 2 | Lasso 3 | Lasso 4 |
|---|---|---|---|---|
| Lasso 2 | 0.94 | | | |
| Lasso 3 | 0.85 | 0.81 | | |
| Lasso 4 | 0.97 | 0.91 | 0.85 | |
| Lasso 5 | 0.99 | 0.94 | 0.87 | 0.99 |

# When could Predictions be Useful?

Tasks with a prediction purpose:

- Predict stock or commodity prices using Twitter data.
- Nowcasting unemployment rate or GDP using Google search queries.
- Pre-screening of job applications.
- Consumer demand (shipping before the order occurs).
- Movie recommendations on Netflix.
- Handwriting, image, face, or voice recognition.

# Examples of Business and Economic Studies

**Prediction Tasks:**

- Chandler, Levitt, and List (2011) predict shootings among high-risk youth to target mentoring interventions.
- Kleinberg, et al. (2018) predict the crime probability of defendants released from investigative custody to improve judge decisions.

**Pre-Processing Unstructured Data:**

- Glaeser et al. (2016) use images from Google Street View to measure block-level income in New York City and Boston.
- Kang et al. (2013) use restaurant reviews on Yelp.com to predict the outcome of hygiene inspections.
- Kogan et al. (2009) predict volatility of firms from market-risk disclosure texts (annual 10-K forms).

# Predictions vs. Causal Inference

- Outcome (e.g., earnings): $Y$
- Binary Treatment (e.g., participation in training program): $D \in \{0, 1\}$
- Potential Outcome:
  - $Y(1)$ potential earnings under participation
  - $Y(0)$ potential earnings under non-participation

  $\rightarrow$ Only one potential earnings can be observed
- Causal effect: $\delta = Y(1) - Y(0)$

$\rightarrow$ Predictions have the observable estimation target $\widehat{Y}$

$\rightarrow$ Causal inference has the (partly) unobservable estimation target $\widehat{\delta}$

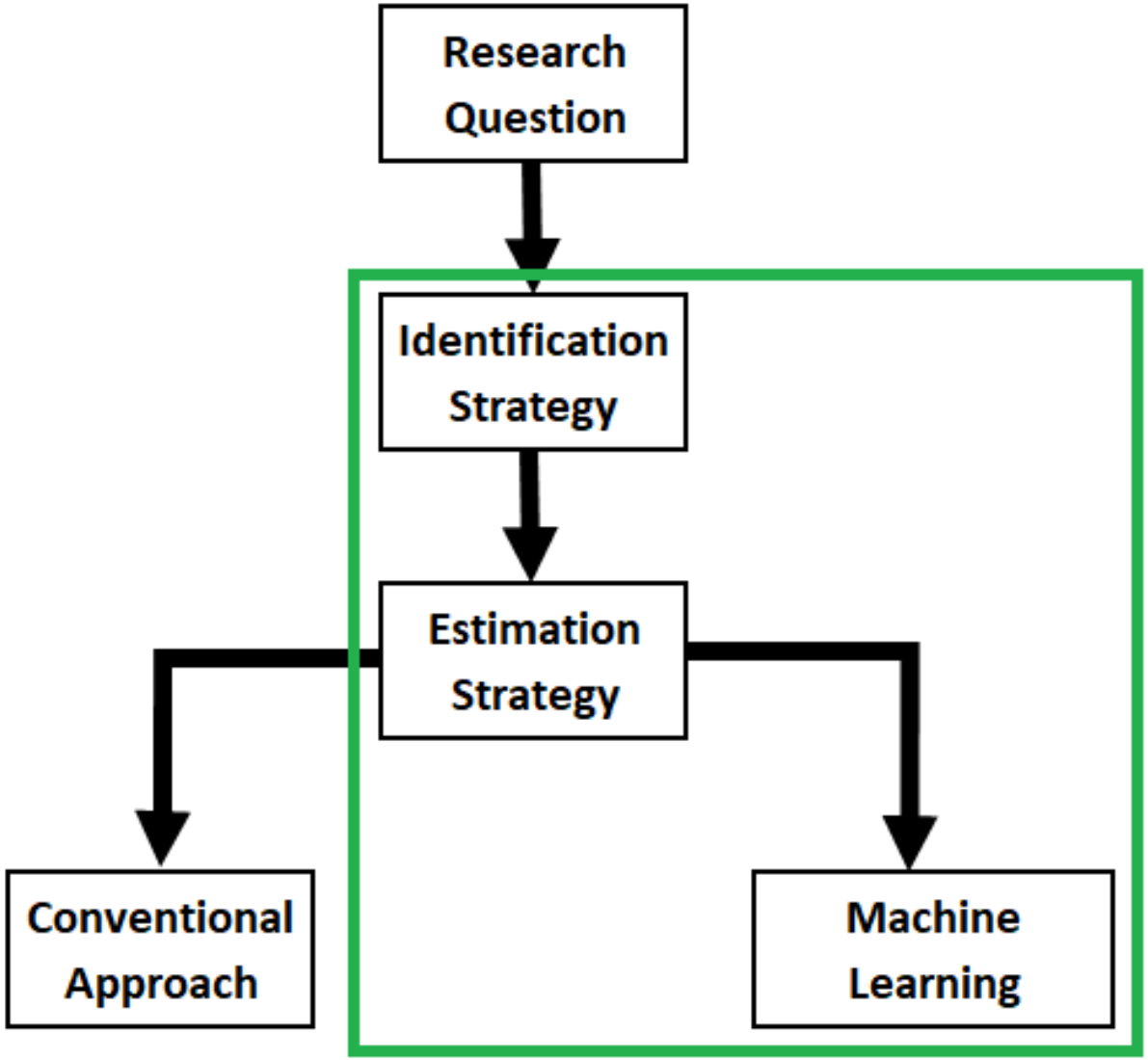
# Training of ML Algorithms

Out-of-Sample Mean-Squared-Error (MSE):

$$MSE_{\hat{\delta}} = E\left[(\hat{\delta} - \delta)^2\right] = \underbrace{E\left[(\hat{\delta} - E[\hat{\delta}])^2\right]}_{\text{Variance}} + \underbrace{E[\hat{\delta} - \delta]^2}_{\text{Squared Bias}}$$

$\rightarrow$ $\delta$ is unobservable

# Research Design



# Potentials of Causal Machine Learning (CML)

**Four potential applications of CML:**

1. Account for (very) many instruments in IV or Heckit approach (prediction problem, issues with inference).
   References:
   - Belloni, Chen, Chernozhukov, and Hansen (2012)
   - Hansen and Kozbur (2014)
2. Account for confounders, e.g., in matching, IV, or difference-in-difference approaches:
   - ML enables the incorporation of (very) many covariates which can make the exclusion restriction more credible.
   - Some ML approaches make little functional form assumptions.

   Reference:
   - Chernozhukov et al. (2017)

# Potentials of Causal Machine Learning (CML)

**Four potential applications of CML:**

1. Account for (very) many instruments in IV or Heckit approach (prediction problem, issues with inference).
   References:
   - Belloni, Chen, Chernozhukov, and Hansen (2012)
   - Hansen and Kozbur (2014)
2. Account for confounders, e.g., in matching, IV, or difference-in-difference approaches:
   - ML enables the incorporation of (very) many covariates which can make the exclusion restriction more credible.
   - Some ML approaches make little functional form assumptions.

   Reference:
   - Chernozhukov et al. (2017)

# Potentials of Causal Machine Learning (CML)

3. Heterogeneous effects:
   - Principled approach makes it less likely to overlook important heterogeneity.
   - Problems: Issues with interpretability and works only for the low-dimensional case.

   References:

   - Wager and Athey (2018)
   - Chernozhukov, Demirer, Duflo, and Fernéndez-Val (2018)
4. Optimal policy rules (e.g. Bandits):
   - Focus on the (discrete) treatment decision instead on the effect size.

   Reference:

   - Athey and Wager (2019)

# Potentials of Causal Machine Learning (CML)

3. Heterogeneous effects:
   - Principled approach makes it less likely to overlook important heterogeneity.
   - Problems: Issues with interpretability and works only for the low-dimensional case.

   References:

   - Wager and Athey (2018)
   - Chernozhukov, Demirer, Duflo, and Fernéndez-Val (2018)
4. Optimal policy rules (e.g. Bandits):
   - Focus on the (discrete) treatment decision instead on the effect size.

   Reference:

   - Athey and Wager (2019)

# Limitations of Causal Machine Learning (CML)

- ML algorithms cannot distinguish between causation and correlation.
  - $\rightarrow$ CML will not select the relevant causal parameters automatically.
  - $\rightarrow$ We have to provide some structure to the CML algorithm.
- CML can estimate causal effects only for a few (usually only one) endogenous variables.
  - $\rightarrow$ We will not obtain the (complete) structural model.
- Identifying assumptions do not change, no matter if we use ML or conventional methods.
- We should resist the temptation to interpret prediction models in a causal way.

# Applications of CML Methods

- Davis and Heller (2017) investigate the effects of summer jobs on the probability to commit a violent crime.
- Taddy et al. (2016) investigate the heterogeneous effects of A/B-experiments in online-auctions (EBay) on customer responses (experimental study).
- Bertrand et al. (2017) and Knaus, Lechner, and Strittmatter (2020) estimate heterogeneous employment effects of training programmes for unemployed persons.